Internet of Things

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Outline

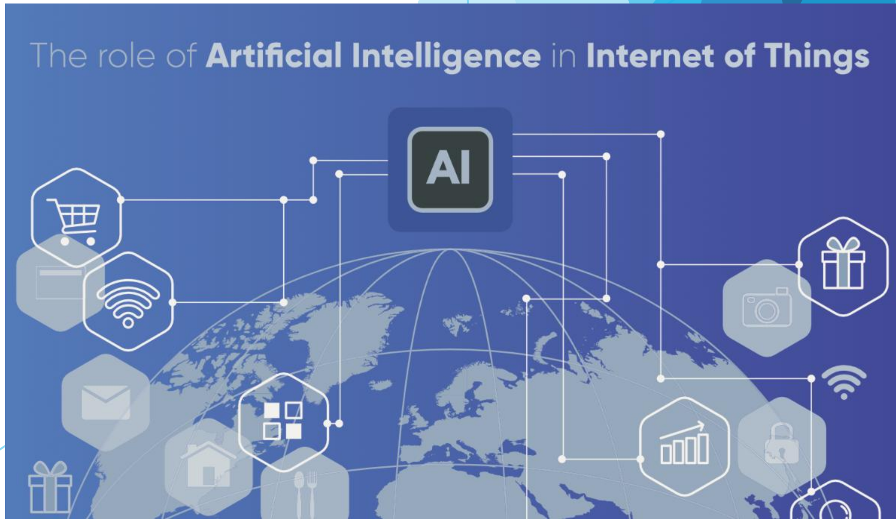
- AI-Driven Internet of Things system
- Components of AI-driven IoT
- Use cases AI-driven IoT
- TinyML
- Applications of AI-driven IoT
- Future improvement in IoT
- Practice code AI-Driven IoT

AI-Driven Internet of Things system

- IoT is like the body, and AI the brains, which together can create new value propositions, business models, revenue streams and services.

AI-Driven Internet of Things system

- The convergence of Artificial Intelligence (AI) and the Internet of Things (IoT) has ushered in an era of intelligent automation, creating systems that can perceive, learn, and react to their environment with minimal human intervention. The Internet of Things, at its core, is a vast network of interconnected devices embedded with sensors, software, and other technologies that enable them to collect and exchange data. The sheer magnitude of this network, with projections estimating billions of connected devices by 2025, generates an unprecedented volume of data, necessitating intelligent processing to derive meaningful value.



AI-Driven Internet of Things system

- Complementing this data-generating capability is Artificial Intelligence, a broad field focused on creating machines capable of performing tasks that typically require human intelligence. Machine Learning (ML), a critical subset of AI, empowers computers to improve their performance on a task through experience, without being explicitly programmed. The integration of AI's analytical prowess with IoT's pervasive connectivity gives rise to AI-Driven IoT (AIoT), a synergy that yields smarter, more efficient, and increasingly autonomous systems. This powerful combination enables real-time data analysis, unlocks predictive capabilities, facilitates personalized experiences, and enhances automation across a diverse range of industries.

AI-Driven Internet of Things system

- The Internet of Things comprises several core concepts, starting with the devices and sensors that serve as the eyes, ears, and hands of the system, collecting data from the physical world or interacting with it based on processed information. The increasing variety and sophistication of these sensors, including advancements in AI-enhanced imaging and the emergence of virtual sensors that use computational algorithms to derive data, are continuously expanding the potential applications of IoT.

AI-Driven Internet of Things system

- LPWAN, and even satellite communications provide the necessary infrastructure for this data exchange. The evolution of these connectivity technologies, particularly the rollout of 5G networks offering ultra-fast speeds and low latency, alongside the development of hybrid connectivity approaches that combine the strengths of multiple technologies are vital for supporting the ever-growing demands of IoT deployments.
- Finally, data processing is the stage where the raw data collected by IoT devices is analyzed to extract meaningful information. This processing can occur at the edge, directly on the device or a nearby gateway, in the cloud, leveraging the vast computational resources of remote servers, or through a hybrid approach that distributes the workload. The increasing importance of edge computing for reducing latency and enabling real-time decision-making is particularly relevant in the context of AIoT applications.

AI-Driven Internet of Things system

- unsupervised learning, where models find patterns in unlabeled data; and reinforcement learning, where models learn through trial and error based on rewards and penalties. Each of these learning paradigms finds relevance in different IoT data analysis tasks. For instance, supervised learning is often used for classification tasks like anomaly detection, where the model learns to identify unusual patterns based on historical examples of normal and abnormal behavior.

AI driven uses

- Regression models, another type of supervised learning, are valuable for predicting continuous values such as temperature forecasts or the remaining lifespan of equipment, providing crucial insights for resource management and predictive maintenance in IoT deployments. Unsupervised learning techniques like clustering can identify inherent groupings in IoT data, which can be useful for segmenting user behavior patterns or categorizing device performance for targeted analysis and optimization. Beyond these fundamental algorithms, deep learning, a subfield of machine learning that utilizes neural networks with multiple layers, has proven highly effective in tackling more complex IoT scenarios such as image recognition in surveillance systems or speech recognition in smart assistants.

AI driven uses

These deep learning models can automatically learn intricate features from vast amounts

of IoT data, enabling increasingly sophisticated AI-driven functionalities.

The intelligence of AI-driven IoT systems is heavily dependent on the quality and availability of data. The process begins with data acquisition, where data is collected from a myriad of IoT sensors and devices using various methods

AI driven uses

- Cloud services, offered by platforms like AWS IoT, Azure IoT, and Google Cloud IoT, provide robust infrastructure for collecting, storing, and managing the large volumes of data generated by IoT deployments. Once collected, the raw IoT data often requires cleaning and preprocessing to ensure its quality and suitability for training AI/ML models

AI driven performance

- Removing noise and outliers, which can skew model training and lead to inaccurate predictions, is another critical step. Data transformation, such as normalization and scaling, ensures that the data is in a consistent format and range, preventing certain features from dominating the model due to their magnitude. Additionally, feature engineering, the process of selecting, transforming, and creating relevant features from the preprocessed data , can significantly enhance the performance and efficiency of AI models in IoT applications by providing the model with more informative inputs. The quality and quantity of this prepared data are paramount, as they directly influence the accuracy and reliability of the AI/ML models.

AI-driven anomaly detection

- AI-driven anomaly detection can significantly enhance the security of IoT networks and enable predictive maintenance by identifying potential failures before they occur. In scenarios where the goal is to predict a continuous value, such as forecasting future temperature trends for smart climate control or estimating the remaining useful life of industrial machinery, regression models are employed.

AI-driven uses

- These models can provide valuable insights for optimizing resource allocation and proactively planning maintenance schedules. Clustering algorithms, which group similar data points together, can be particularly useful in IoT applications for identifying patterns in user behavior within smart homes or analyzing device performance across a large fleet. This can lead to personalized experiences or the identification of common failure modes. Training these AI/ML models with IoT data typically involves a structured process.

standard ML models

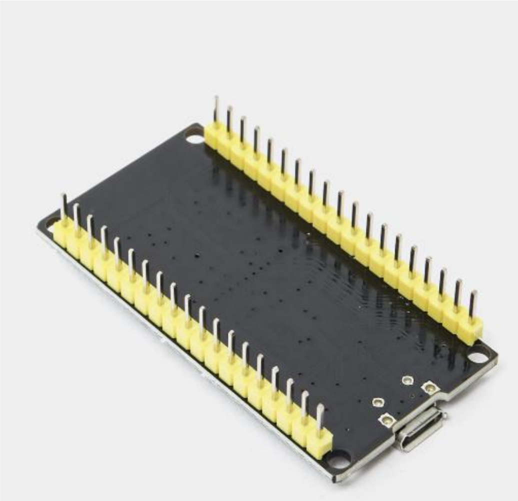
- Why standard ML models cannot run on microcontrollers
- Memory and power constraints

AI-driven uses

- First, the prepared data is divided into a training set, which the model learns from, and a testing set, which is used to evaluate the model's performance on unseen data. Then, an appropriate model architecture is selected based on the task and data characteristics. The model is then trained using the training data, and its performance is iteratively evaluated and refined using the testing data until satisfactory results are achieved.
- This process can be facilitated by various platforms and frameworks, including open-source libraries like TensorFlow and scikit-learn, as well as cloud-based machine learning services that offer scalable compute resources and pre-built algorithms.

Optimizing and deploying AI models

- To bring intelligence directly to the point of data generation, deploying AI models on IoT devices themselves, a concept known as Edge AI, is gaining significant traction. This approach offers several compelling advantages, including reduced latency in decision-making, enhanced privacy by keeping sensitive data on the device, and the ability for IoT systems to function reliably even when network connectivity is intermittent. A specialized area within Edge AI is TinyML, which focuses on optimizing and deploying machine learning models on extremely resource-constrained microcontrollers.



ESP32

AI capabilities on low-power devices

- TinyML enables AI capabilities on low-power devices such as Arduino and ESP32, opening up a vast array of embedded AI applications. Resources are available to guide developers through the process of deploying AI models to specific hardware platforms. For Raspberry Pi, frameworks like TensorFlow Lite can be used, and tools like Ollama facilitate the deployment of Large Language Models (LLMs) on these small yet powerful boards.

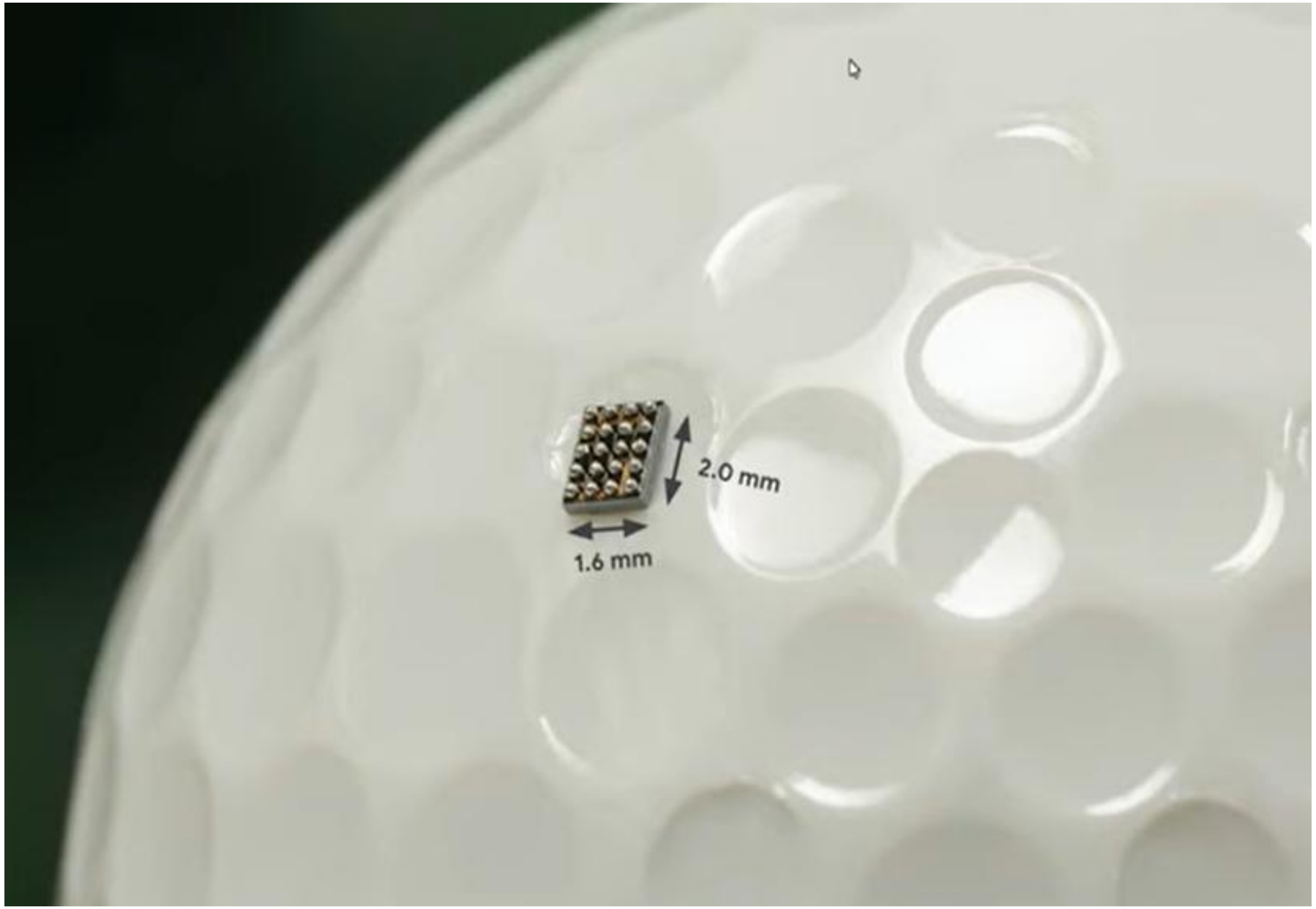
RAM: KBs
No GPU
Real-time inference required

CPU size

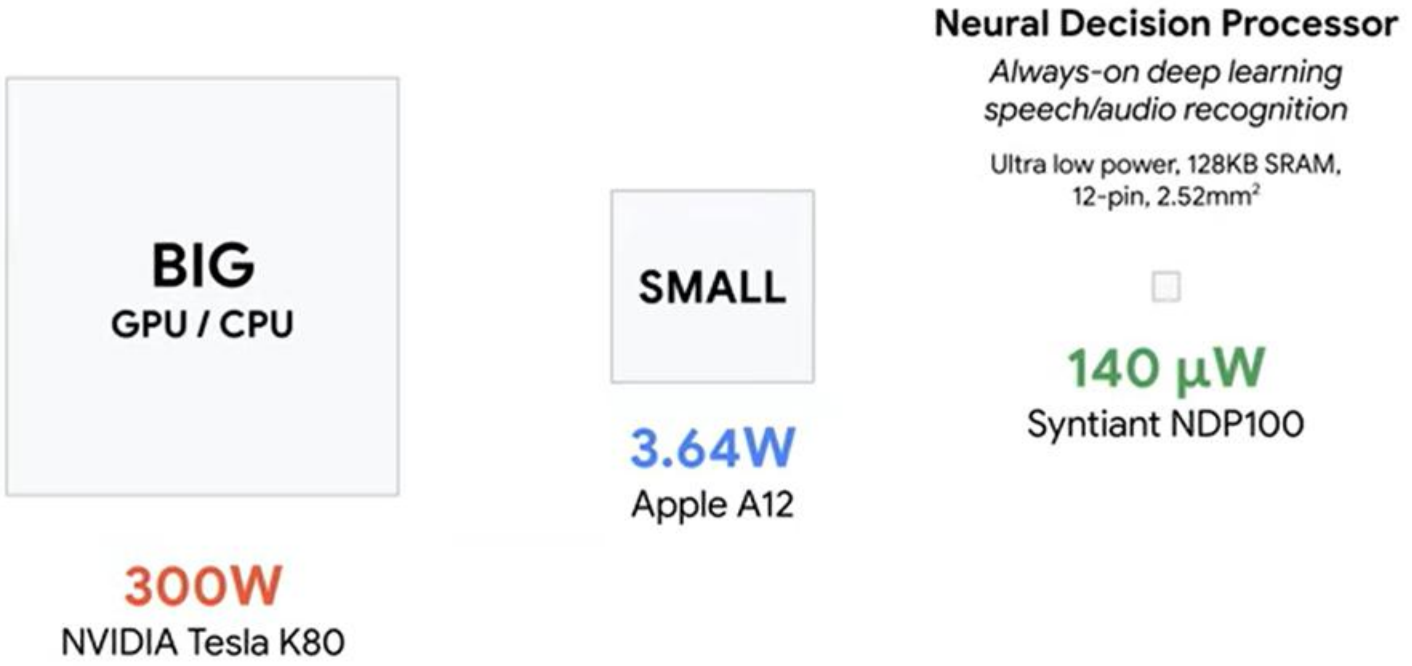
Smart watch processor

Enable TinyML

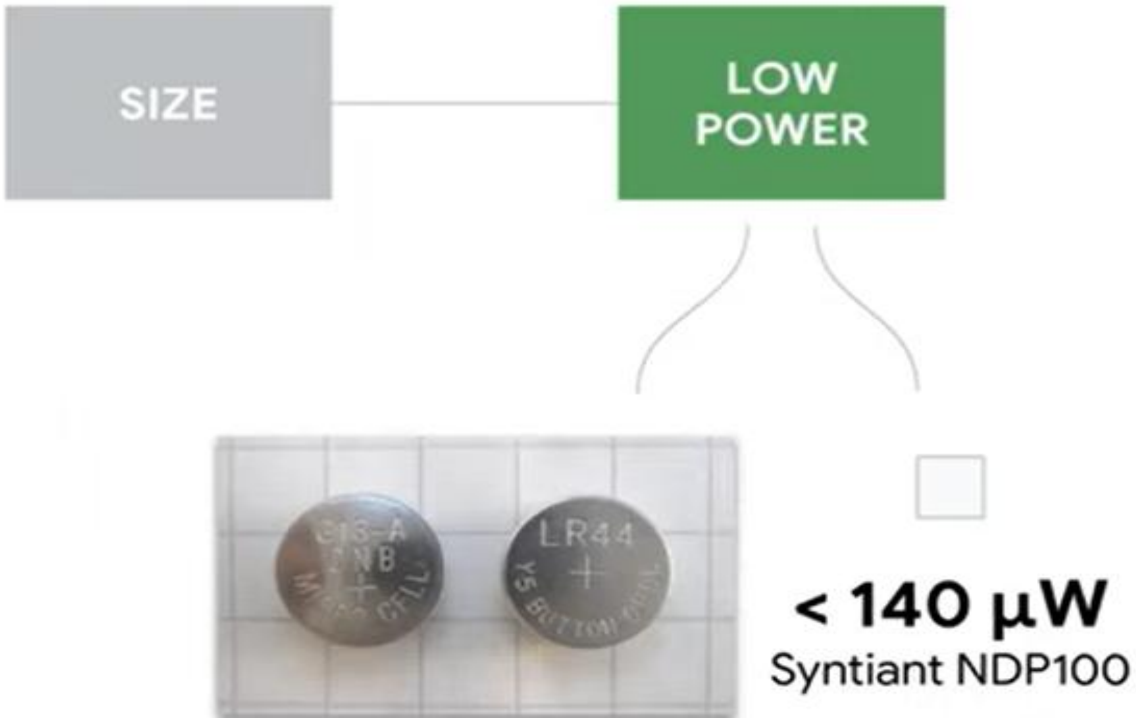
Enable TinyML



Comparing power



TinyML enables ML inference on resource-constrained microcontrollers (MCUs)



Microprocessor vs Microcontroller

Heart of computer system	Heart of embedded system
Just the processor, memory and storage are external	Memory and storage are all internal to the system
Mainly used in general purpose system like laptops, desktop, servers	Mainly used in specialized, fixed function system like phones, MP3 players etc
Offers flexibility in design	Limited flexibility in design
System size is big	System size is tiny

Key Difference

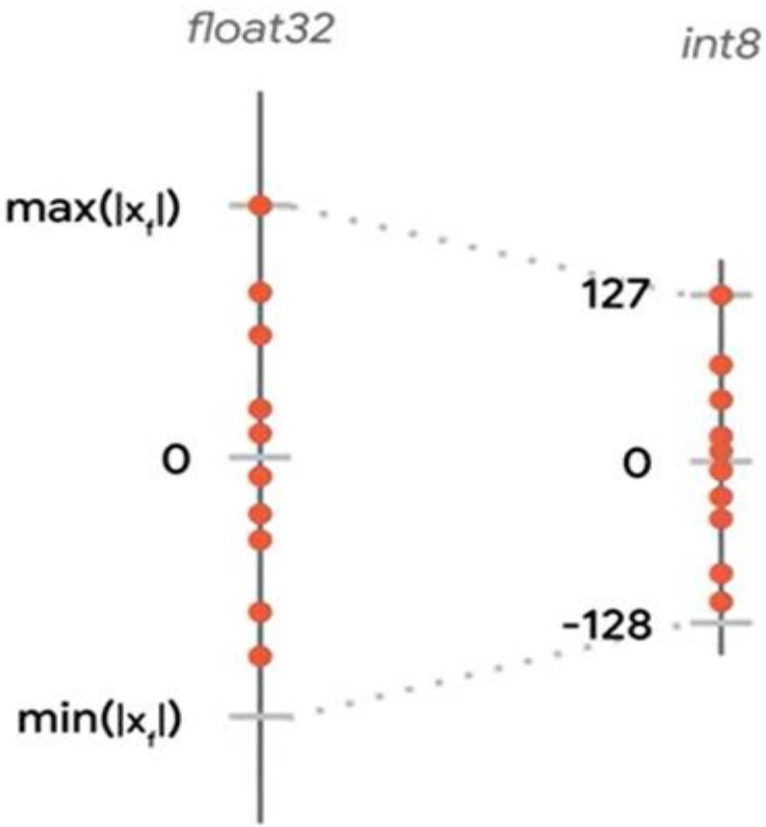
	Microprocessor	>	Microcontroller
Platform			
Compute	1GHz–4GHz	~10X	1MHz–400MHz
Memory	512MB–64GB	~10000X	2KB–512KB
Storage	64GB–4TB	~100000X	32KB–2MB
Power	30W–100W	~1000X	150µW–23.5mW

- How complicated is the running task?
- How much memory does it need to have?
- How long does the job have to perform?

Key Difference

	TensorFlow	TensorFlow Lite
Topology	Variable	Fixed
Weights	Variable	Fixed
Binary Size	Unimportant	High Priority
Distributed Compute	Needed	Not Needed
Developer Background	ML Researcher	Application Developer

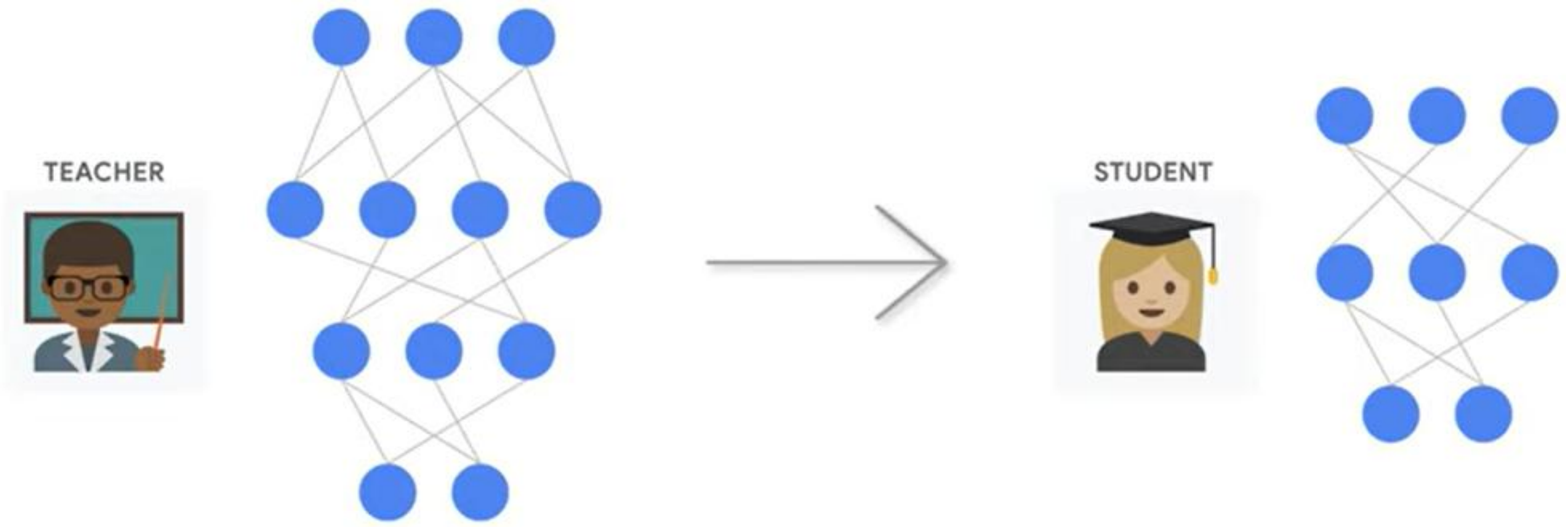
Quantization



Reduce precision: 32-bit → 8-bit
Smaller and faster models

Quantization is a model compression technique that reduces the numerical precision of a model's weights and activations. Most models are trained using high-precision **32-bit floating-point** (FP32) numbers; quantization converts these into lower-precision formats like **8-bit integers** (INT8) or 4-bit formats

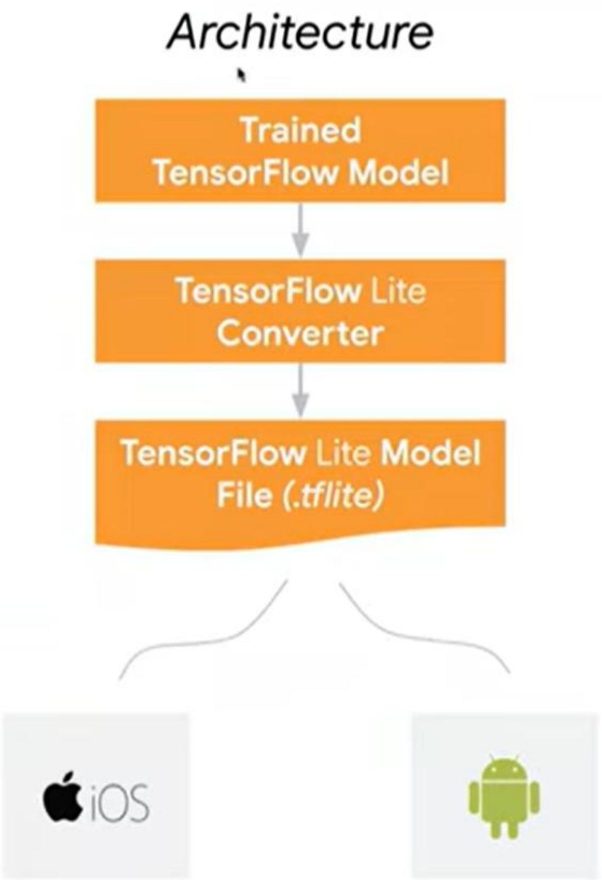
Knowledge distillation



Train small model from large model
Maintain performance

(e.g., DistilBERT achieves ~97% of BERT's performance with 40% less size).

Conversion



Application:
Keyword spotting
Activity recognition
Smart sensors

Pruning

- Remove unnecessary weights
- Structured and unstructured pruning

Efficient Architectures

- MobileNet, TinyCNN
- Depthwise separable convolution

Compression

- Weight sharing
- Encoding techniques

Pipeline

Train → Optimize → Convert → Deploy

AI capabilities on low-power devices

- Arduino, known for its simplicity and low power consumption, can also host AI models using libraries like EloquentTinyML, which simplifies the conversion of trained models into C code suitable for microcontrollers. ESP32, with its integrated Wi-Fi and Bluetooth, is another popular platform for AIoT projects, with frameworks like TensorFlow Lite for Microcontrollers and end-to-end platforms like Edge Impulse streamlining the development and deployment process.

Practical use cases

- AI can be used to control lighting, thermostats, and appliances based on sensor data and learned user preferences. For instance, a smart thermostat can learn a user's daily schedule and automatically adjust the temperature for optimal comfort and energy efficiency. Environmental monitoring is another area where AIoT excels. By deploying sensors to track parameters like temperature, humidity, and air quality, and then leveraging AI for analysis, systems can identify patterns, detect anomalies like pollution spikes, and trigger alerts to inform users or authorities.

Case Study

- Keyword spotting model (~20KB)
- Audio → MFCC → NN

Challenges

Accuracy vs size trade-off
Memory constraints

Applications of Artificial Intelligence(AI) in Internet of Things(IoT)

- Collaborative Robots
- Drones
- Smart Cities
- Digital Twins
- Smart Retailing

Applications of Artificial Intelligence(AI) in Internet of Things(IoT)

- **Collaborative Robots**
- Cobots are highly complex machines that are designed to help humans in a shared workspace with environments ranging from office to industrial.
- **Drones**
- navigate unknown surroundings such as offshore operations, mines, war zones or burning buildings
- **Smart cities**
- energy efficiency, air pollution, water use, noise pollution, traffic conditions, etc.

Applications of Artificial Intelligence(AI) in Internet of Things(IoT)

- **Digital Twins**
- Digital Twins are mainly used to analyze the performance of the objects without using the traditional testing methods and so reducing the costs required for testing.
- **Smart retailer**
- AI and IoT can be used by retailers to understand the customer behavior (by studying the consumer online profile, in-store inventory, etc.) and then send real-time personalized offers while the customer is in the store.

Summary

- The synergy between AI and IoT unlocks a vast potential for innovation across numerous sectors, from enhancing everyday convenience in smart homes to driving efficiency and safety in industrial and environmental applications. The ability to collect, process, and analyze data intelligently, whether at the edge or in the cloud, empowers the development of increasingly sophisticated and autonomous systems. As the field continues to evolve, advancements in hardware, software, and connectivity will undoubtedly pave the way for even more transformative AIoT solutions.

Summary

- Tiny transformers
- On-device learning

- TinyML = AI on microcontrollers
- Efficient and low power

Practice code AI driven IoT

- Enhance the DataProcessor class to incorporate more sophisticated AI techniques.

  Instead of a simple RandomForestRegressor, experiment with other machine learning models from scikit-learn, such as:

- **Support Vector Regression (SVR)**
- Evaluate the performance of the different models using appropriate metrics (e.g., Mean Absolute Error) and compare the results.

Reference

- https://www.geeksforgeeks.org/the-role-of-artificial-intelligence-in-internet-of-things/

Question

Thank you